

Genotype x environment interactions and local adaptation

Learning objectives

1. Understand the difference between GxE, phenotypic plasticity, and local adaptation
2. Know how we detect each of these processes

last lecture: Introduced multiple populations but assumed all individuals in population react to environment in same way

several ways plant responses to environment vary within a population

Genotype x environment interactions

Phenotypic plasticity vs. GxE

- Phenotypic plasticity - traits shift with environment within a genotype
- GxE - trait or fitness shifts with environment depend on genotype *different genotypes have different responses*

Plot E x trait with different lines for different genotypes

plot1: No slope - no phenotypic plasticity, no GxE (what we assumed prior to this)

plot2: Parallel lines with different intercepts - plasticity without GxE

plot3: Slopes differ but don't cross - plasticity with variance-changing GxE

plot4: Slopes differ and cross - plasticity with rank-changing GxE

Methods

- Common gardens - genotypes planted in same environment
- Reaction norms - plant response to a range of environmental conditions *what we drew above*

Local adaptation

main difference between this and the others is this has to do with fitness, not trait shifts

- “adaptive” = affects fitness
- environment -> phenotype -> fitness
- local adaptation - fitness highest in current environment *traits adapted to local conditions*

Plot environment A & B by fitness

Plot1: Local adaptation - a line high in A, low in B; b line high in B low in A

plot2: One species more adapted - a line higher than B line, either increasing or decreasing between A and B

plot3: Maladaptation - opposite of local adaptation

Methods

- Reciprocal transplant or common garden *local and non-local plants in same condition*
- Reaction norms
- fitness highest in original/source conditions